DESIGNING OF AN ASSISTIVE ROBOT FOR WASTE MANAGEMENT

by

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DESIGNING OF AN ASSISTIVE ROBOT FOR WASTE MANAGEMENT

DESIGN PROJECT REPORT

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ABSTRACT

This project presents the design and development of an autonomous waste management robot prototype built using LEGO SPIKE Prime and OpenMV H7 hardware. The robot, Waste bot, is engineered to independently navigate its environment, identify waste objects, and transport them to their corresponding disposal bins without human intervention. This project seeks to contribute to a circular and sustainable environment through autonomous waste management.

The system integrates the LEGO SPIKE Prime platform for locomotion and mechanical actuation with the OpenMV H7 microcontroller's onboard machine vision capabilities to perform real-time waste detection and classification. Utilizing the edge impulse platform and small datasets, images of plastic, metal, and paper waste are trained to create a machined learning model. This waste classification model is deployed on the OpenMV H7. The robot identifies waste items by category and delivers them to appropriate bins which are identified using April Tags. The Lego Spike Prime hubs reads data from a distance sensor, to then drive motor movement for navigation. This enables the robot to traverse its environment, avoid obstacles, and execute precise pickup-and-deposit actions.

The machine learning model is constrained in size due to OpenMV camera’s limited hardware. This results in lower accuracy, especially in metal and plastics as they share similar features. After 20 test trials for each waste class, the waste classification model reveals an accuracy of 81%.

This prototype demonstrates the feasibility of combining accessible educational robotics hardware with embedded machine vision to address practical environmental challenges. The project contributes to the growing field of intelligent waste sorting automation, with implications for scalable, low-cost solutions in waste management contexts ranging from classrooms and offices to broader public spaces. Future work may explore expanded waste classification datasets, improved navigation algorithms, and enhanced gripper mechanisms to increase sorting accuracy and operational reliability.

DEDICATION

I dedicate this project to my parents, Bara Niang and Anta Kebe, who, through their financial and moral support, became the source of inspiration in my attaining an education.

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This project was written under the direction and supervision of Dr. Abram Musinguzi. I would like to express my sincere appreciation to him for the interest and assistance given to me.

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CHAPTER ONE: INTRODUCTION

1.1 Background

Machines have been given the ability to sense, think, and act in the real world through robotics. Robotics is a multidisciplinary field that combines mechanical, electrical, and computer science principles to create autonomous machines. It is a combination of hardware and software that defines the characteristics of a robot. Hardware can include sensors, and actuators, while software delves into programming, machine learning, and AI. With these tools, a variety of robots can be made which range from rigid arms to collaborative robots, to adaptive intelligent beings. These machines gather information and manipulate their environment based on said information.

The application of robotics is found in many industries due to the cost cutting, time saving, and convenience that they provide. For example, manufacturers utilize pick and place robots to streamline their production processes. In electronic manufacturing spaces, robots are used to place tiny components precisely onto circuit boards [1]. automotive manufacturers utilize pick and place arms to install heavy car parts. pharmaceutical companies employ robots to operate the creation and packaging of pills as they eliminate contamination [1]. Currently, robotics is found mostly in logistics and manufacturing areas. Although there are not many examples found in the home, a notable implementation includes cleaning robots like Roomba [2]. Furthermore, the pick and place capability of robotics is suitable in multiple circumstances. One vital use would be for the handling of excess waste in the world.

1.2 Problem Statement

Every passing day, humans are contributing to the global crisis that is waste pollution. This has led to an annual generation of two billion tons of waste, which is expected to reach 3.78 billion tons by 2050 [3]. The improper disposal and buildup of trash, food, plastic, paper, agricultural, textile and industrial waste causes abundant climate, health, and environmental problems [3]. One consequence is revealed in a 2021 research that claims 19 million tons of plastic waste enter aquatic environments annually, where marine life consumes it [4]. Additionally, the neglection of waste releases dangerous chemicals which are known to cause cancer, infertility, and Alzheimer's, and many other health complications. The persistence of waste pollution because of poor waste management could lead to the downfall of humanity.

The necessary improvements for reducing waste pollution include a better waste management and resource recovery system. The current initiatives of reaching these goals can be studied at an individual, governmental, and societal level. In terms of waste management, policies are being proposed to crack down on single use plastic [5]. Additionally, a recent report proposes a smart waste management system for a city of Johannesburg, which monitors dumping sites and bins for proper disposal [6]. Another solution being provided is by a company called Waste Robotics. The company utilizes robot arms and conveyor belts to sort garbage collected at sites [7]. On the other hand, Resource recovery, in terms of plastic, is failing as a 2019 report reveals only 9 % of global plastic is recycled [8]. Fortunately, efforts are being made to raise societal awareness. One such example is annual international zero waste day in March. Overall, current efforts of

fighting waste pollution include global monitoring, employing robotics, raising awareness, policy changes, and city infrastructure changes.

To tackle the growing waste pollution, we developed a mobile robotic system which would automate the collection of waste and recyclable material. This robotic system has the capability to detect and categorize waste into plastic, metal, or trash and place it into its dedicated bin. With a mobile pick and place robot implemented in city environments, waste can be detected in its early stages and classified correctly. Furthermore, the mobile robot provides a substitute for the neglect of proper waste disposal and a cleaning function.

1.3 Objectives

The main goal of this capstone project is to provide a solution for waste pollution by designing and developing a mobile robot that can autonomously identify, collect and sort waste into categories of trash, metal, and plastic

The specific objectives are

1. To design the pick and place robot
2. To build the pick and place robot
3. To program pick and place robot using python
4. To test the multiple functions of the pick and place robot

1.4 Requirements

Functional Requirements

- The system shall navigate its environment while avoiding obstacles
- The system shall classify waste correctly (metal, plastic, trash)

- The system shall pick and place objects of varying shapes
- The system shall transport waste materials to designated bins
- The system shall contain a user interface for control (mobile app, onboard, display)

Non-functional Requirements

- The system shall detect waste at success rate of 80% or higher
- The system shall be lightweight
- The system shall contain user-friendly controls
- The system shall safely navigate without causing harm to animals and objects
- The system shall operate for a prolonged amount of time

1.5 Significance of Project

Table 1: Significance of Project

Academic	Provides a guide to developing a robotics system. Additionally, it provides a tutorial on the integration of certain equipment and the capabilities of low-cost equipment in robotics.
Industrial	Provides an application of robotics outside manufacturing and logistics. Furthermore, it provides an alternative to labor costs required to collect recyclable materials.
Economic/social	Showcases the capability of robotics in environmental services. If implemented on a larger scale, it also may reduce costs, as less material must be produced. The mobile pick and place robot also provides an alternative to human labor

1.6 Scope of Project

This project will cover the design and building of small-scale mobile pick and place robots.

It also covers the programming of obstacle detection, picking and placing and waste

classification. Finally, tests must be done to see if the robot functions as intended and reacts to certain environments. The scope of the project is to build a prototype.

The project will not cover industrial scale implementation. In addition to this, certain waste such as glass, hazardous, and biodegradable waste will not be included in the system. The project will not cover the material recycling process. The robot will not be tested in uneven terrain or with waste items of large sizes and heavy weight. The project will not cover commercialization.

1.7 Project Limitations

Budget: Due to budget limitations, the main equipment being utilized is the Lego Spike Prime kit along with third party components.

Weight: The Lego Spike equipment only allows picking and placing objects that weigh less than 100 grams.

Size: With the equipment used and the small scale of the robot, the grippers can only transport objects equal or smaller size. Examples include water bottles, brush, and a small notebook. Additionally, the small scale of the robotics system proves difficult for reaching objects at an elevated level.

Power: The hub provides power for every component in the system with the use of a 2100mAh rechargeable battery. Considering the power consumption of the motors and the added microcontrollers, operating time may be limited. This requires recharging the system from time to time.

Outdoor Navigation: Navigation in an outdoor environment may prove difficult, due to uneven terrain, and unpredictable obstacles.

Skill: At the current level of skill and equipment, a fully autonomous system is not possible. Therefore, the robotic system will locate bins utilizing fiducials tags.

1.8 Capstone Chapter Designation

The subsequent chapters are structured in the following format. **Chapter 2** delves into the gathering of relevant projects, research, and resources for creating the robot. **Chapter 3** is the designing of the robot, which includes gathering necessary software and hardware components to build system architecture. **Chapter 4** involves final design and analysis in which hardware assembly, programming, and testing of the system takes place. Finally, **Chapter 5** will discuss the conclusion of the project and recommend improvements.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The field of robotics has seen a recent increasing trend in utility from manufacturing to healthcare to space exploration and even in the home. An advantageous use of robotics would be to employ it in tackling the growing global crisis of waste pollution. When designing a Lego waste management pick and place robot, many challenges occur when processing, drawing, building, and coding such a complex system. The methodology of building a machine requires time, effort and most importantly, research. Specifically, how the field of robotics developed which pertains to serving the creation of a pick and place Lego waste management robot.

It is important to study the history of robotics because of what we can learn from past projects and their methodology. Research on the past applications of robotics provides insight into developing an autonomous waste management system. Therefore, the goal of this review is to analyze relevant past robotics projects to address gaps needed to complete a waste management robot.

2.2 Evolution of Robotics

Robotics systems, in some form, are found throughout history. The methodology for creating autonomous machines continually improves. Throughout history, automated systems have been used to improve the quality of work, health, sustainability, and many other aspects of life. One can say, elements of robotics can even be found in ancient civilizations. Examples would include mechanical contraptions made for lifelike mimicry

and convenience [9]. Further advancements in robotics occurred in the 20th century. The creation of Unimate in 1954 marked the first autonomous industrial robot for welding and material handling [9]. The next key milestone would be the introduction of microprocessors and microcontrollers in the early 1980s. These tools would lead to more autonomous robots being introduced to the consumer market. For example, the first robot vacuum cleaner, the Electrolux Trilobite, became available to consumers [10]. Nowadays, artificial intelligence and machine learning are the new wave of tools implemented into robotics. An example of this is the emergence of self-driving cars, which monitors surroundings on the road and adjust accordingly [11]. In terms of methodology, robotics has advanced from mechanical devices to electrically automated systems. Currently, artificial intelligence and machine learning allow for more independent and sophisticated systems.

2.3 Applications of Robotics

2.3.1 Manufacturing

Robots are integral to manufacturing processes as they improve efficiency, provide consistent quality, and reduce costs. Additionally, manufacturers may include Cobots, collaborative robots, into their processes with humans. Cobots are equipped with safety features to work alongside humans, furthermore, enhancing production capabilities.

A common robot in manufacturing settings is the six degrees of freedom robot arm. Their multiple joints along with interchangeable end effectors allow for the flexibility to paint, weld, handle materials, and many more tasks. Another example is the SCARA robots,

which is a small-scale material handling robot designed for high-speed assembly. They are commonly utilized by electronics and automotive manufacturers [12].



Figure 1: Robotics in plastic manufacturing process [12]

Robots are found in the production processes of Automotive, Electronic, and Plastic Manufacturers. The automotive industry employs robotics for welding, assembling, painting, and inspection [12]. Electronic manufacturers use high speed robots to place tiny components onto circuit boards [12]. In plastic production, Robots are widely used for the injection molding process [12].

2.3.2 Robots in Healthcare

There are many identifiable robots which serve different roles in assisting in healthcare. Some robots made to assist in healthcare include surgical robots and delivery and transport robots. A notable surgical robot is the Da Vinci Surgical system. This is a system which provides instruments controllable through a console to perform minimally invasive surgery [13]. The system has been tested on 103 patients [13]. Another surgical robot is the

ISR'obot, which is designed to assist in visualization and needle guidance in prostate biopsy. Additionally, a robot named TUG was used to transport drugs from a hospital's pharmaceutical department to its intensive care unit [13].



Figure 2: Surgeon operating Da Vinci Surgical System [20]



Figure 3: TUG robot hallway navigation [21]

2.3.3 Domestics Robots

It can be stated that robots are less frequent outside of business settings, a domestic robot which can be found in many homes is Roomba. Roomba is a robotic vacuum cleaner, which navigates a home serving a cleaning function [14].



Figure 4: Roomba Combo detects dirt [22]

2.4 Relevant Design Techniques

2.4.1 Conceptual Design

The process of building a mobile pick and place sorting robotics system involves mechanical design, electronics integration programming, and testing. Most importantly, a conceptual design, which maps out the system and its components, should be the first step. A conceptual design of low-cost arm which segregates waste is shown in Figure 5 [15]. The block diagram maps the process of sensing, controlling, and actuating.

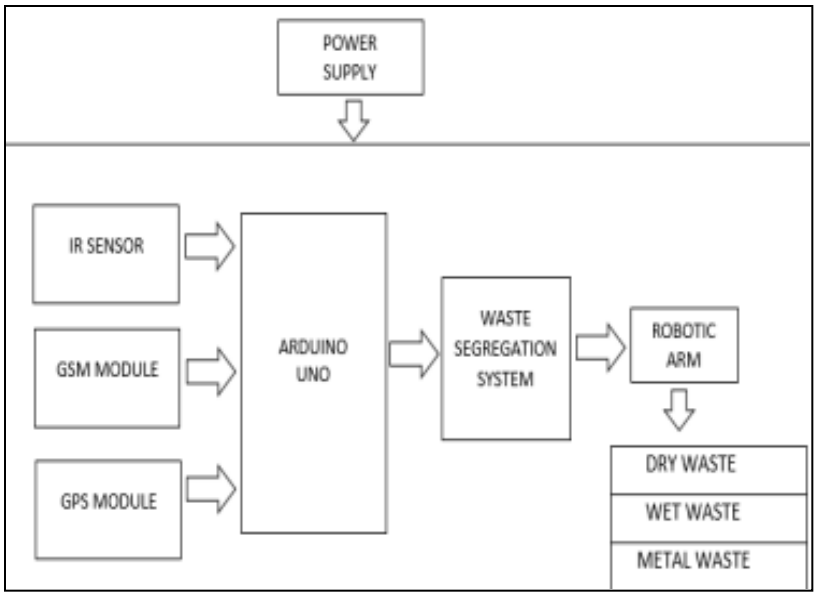


Figure 5: Block diagram of Waste segregator components [15]

2.4.2 Mechanical Design

Mechanical principles can be utilized to create a robotic arm which functions efficiently. In Figure 6, a three-pronged gripper, made with Lego parts is shown [16]. The entire mechanism can be open and closed using one motor, which is made possible through gears.

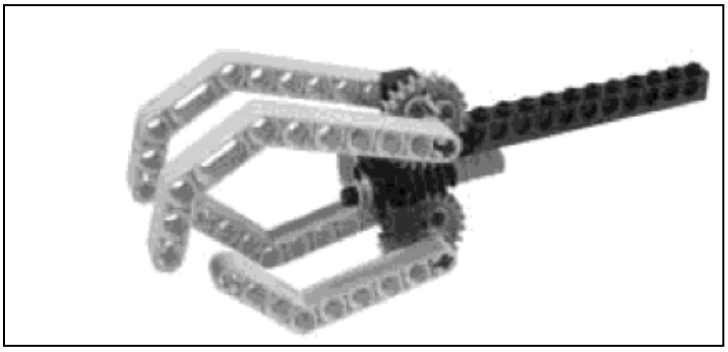


Figure 6: Three-pronged Lego gripper [16]

2.4.3 Sensing Methodologies

A robotics system needs information from its environment to function properly. In the case of waste classification, the multiple approaches to material sensing include color sensors, metal sensors, or a vision-based system. To construct a wall following robot, the report [17] utilizes a Lego robotics kit along with the sensors shown in Figure 7. This allows for a mobile driving base that detects and adjusts to a wall accordingly.

Name	Touch Sensor	Ultrasonic Sensor	Color Sensor	Gyro Sensor
Mechanism Components with Sensors				

Figure 7: Lego Wall following robot components [17]

2.4.4 Programming

Information gathered from sensors are utilized for conditionals in programming. These conditionals dictate the robots' actions. In Figure 8, a flowchart of a localization and navigation algorithm is shown [18]. The algorithm contains a loop which instructs the system to move forward until the ultrasonic sensor detects an object. Then the system path plans and continues forward until another object is detected. The two conditionals in the program include whether an object is detected and whether the destination is reached.

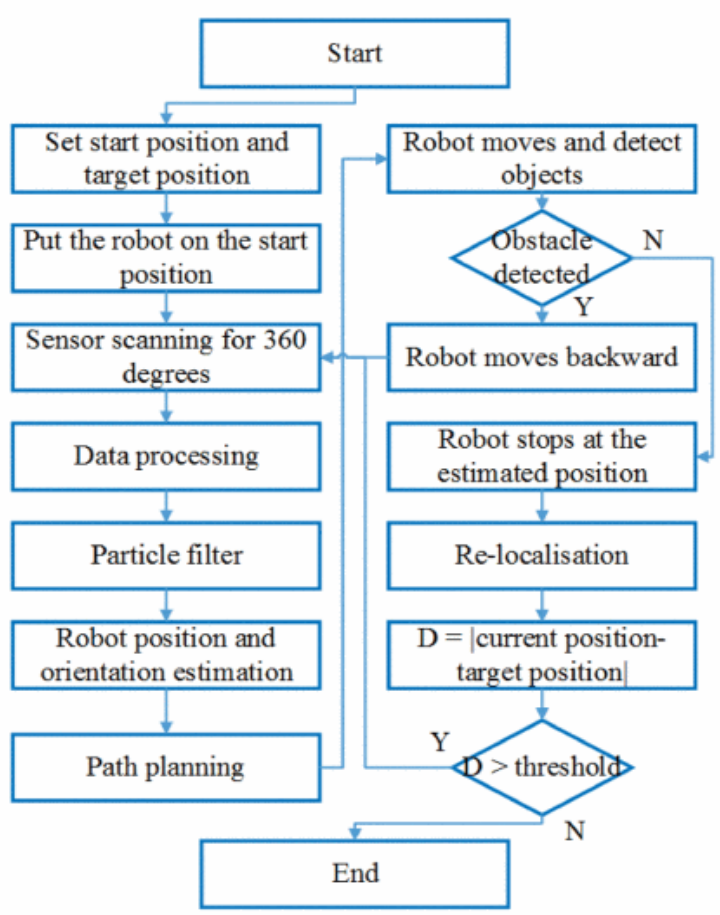


Figure 8: Obstacle detection and path planning flowchart [18]

2.4.5 Testing/ Analysis

Testing robot functionality is crucial. During testing phase, results should be noted as they can detail further improvements needed in a robotics system. Furthermore, the visualization of gathered data could illustrate the system's inadequacies. In Table 2, test cases for the object detection and path planning algorithm are shown below [18]. Additionally, Figure 9 shows a graph revealing accuracy of image classification algorithms in a robotics system with a vision sensor [19].

Table 2: Test result for path planning algorithm [18]

Map	Completion time (s)	Distance from target (cm)	Path length (cm)	Detected collision
Map1	15.22	1.79	116.9	0
Map2	16.69	1.83	120.1	0
Map3	23.48	2.11	276.2	0

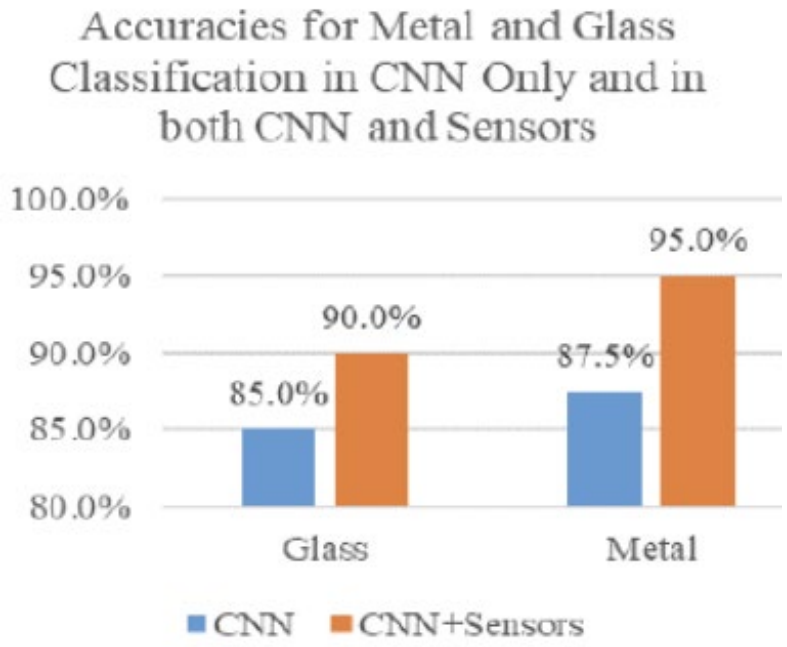


Figure 9: Vision and classification algorithm result [19]

2.5 Summary

This chapter explores the history of robotics, and its applications for understanding the evolution of the field. Then, research was done into relevant projects and their approaches to designing, assembling components, and programming a robotics system. Additionally, key challenges such as programming an obstacle avoidance, and material classification system and integrating were addressed. Furthermore, research into gathering waste datasets for machine learning waste types is necessary; however, an alternative to this is to utilize sensors. Fortunately, these challenges are to be solved in the implementation phase.

CHAPTER 3: DESIGNING AND IMPLEMENTATION METHODOLOGY

3.1 Introduction

In designing an automated waste management system, it is important that it fulfills key functions. These functions can be broken down into functional and non-functional requirements. With the requirement listed, an automated perception to action cycle is then approached through the utilization of components and methodologies.

3.2 Design Requirements

3.2.1 Functional Requirements

- **The system shall navigate its environment while avoiding obstacles in search of waste**

For the task of navigation, a driving base with two wheels is devised. Positioned in its front is a distance which redirects the driving base in the case of an obstacle detected. The waste management system is to blindly roam its environment, until the vision sensor detects waste.

- **The system shall classify waste correctly (metal, plastic, trash)**

Within the waste management system contains a vision sensor for the detection and classification of waste. This vision sensor is the OpenMV H7 camera which will utilize a machine learning algorithm for classifying waste. The vision sensor will also provide coordinates to said waste.

- **The system shall pick and place objects of varying shapes**

A one degree of freedom arm is to be placed above the driving base. The arm contains a gripper for picking waste. The gripper will also contain a force sensor for confirming if the waste is firmly held in place. When designing the arm, it is important to consider the maximum load it must carry and a gripper that suits a variety of items.

- **The system shall transport waste materials to designated bins**

The transport of waste requires the vision for mapping and guiding to the bins. An approach to tackling this problem is to attach RFID tags to the bins, which the sensors can pick up.

3.2.2 Non-Functional Requirements

- The system shall detect waste at success rate of 80% or higher
- The system shall be lightweight
- The system shall contain user-friendly controls
- The system shall safely navigate without causing harm to animals and objects
- The system shall operate for a prolonged amount of time

3.3 System Components

Figure 10 below shows a sketch of the waste management system. It contains a driving base with two wheels along with a vision, sensor, force sensor, and distance sensor. The vision sensor is placed at an elevated level so that its view isn't obstructed. Along the joints of the arm and inside of wheels is a DC motor which creates mechanical movement for the system.

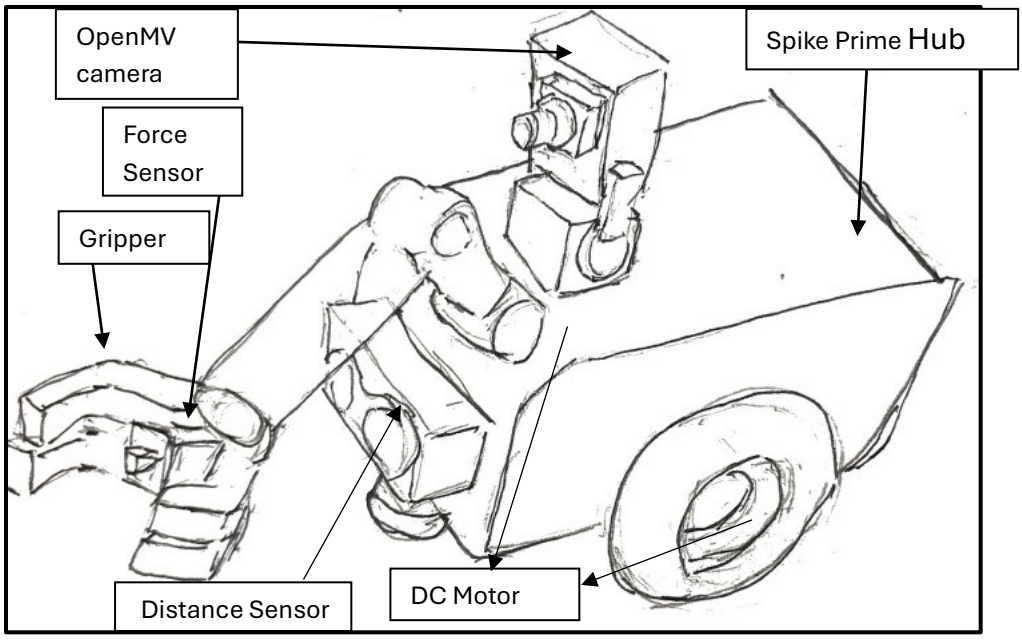


Figure 10: Sketch of The Waste Management System

3.3.1 Spike Prime Kit



Figure 11: Spike Prime Kit [23]

The Lego Spike Prime kit version 143 shown above provides a simple introduction to robotics. The kit contains a hub, DC motor, sensors, and assembly parts made in Lego such as beams, wheels gears, connectors, and bricks. The sensors in the kit include force,

distance, and color sensor. Connecting motors and sensors to the hub only requires plugging them in. In addition to the hub powering everything in the system, the Spike Prime kit makes for easy implementation of robotics systems.

3.3.2 Spike Prime Hub

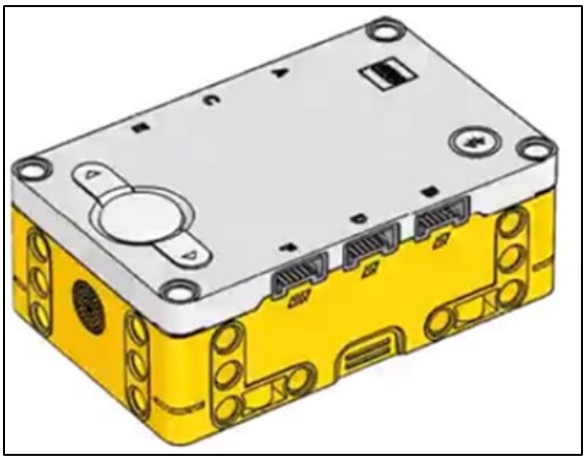


Figure 12: Spike Prime Hub [23]

The Spike prime hub acts as the microcontroller for the Lego robotics system. The hub contains six ports for powering, communicating, and controlling other electrical devices. Controlling the hub can be done with Lego block coding or micro python. Additionally, the hub also contains Bluetooth connectivity, two customizable buttons, and a built-in gyro sensor. The Spike prime hub's main function is as the control center for the waste management system.

3.3.3 OpenMV H7 Camera

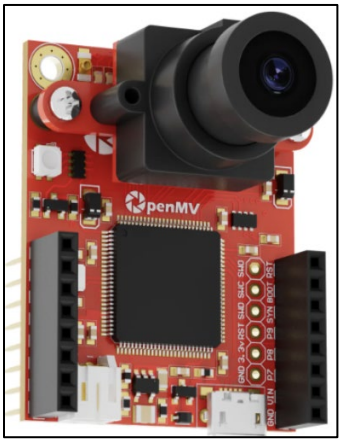


Figure 13: OpenMV H7 Camera [24]

The OpenMV camera is a computer vision microcontroller mainly used for image algorithms. Similar to spike prime hub, the OpenMV camera is coded using micro python. Some demonstrated features of the OpenMV camera includes barcode detection, object, detection, Gesture detection, and Face detection. With the use of a custom breakout board, the OpenMV camera can interface with the Lego hub. Then with the use of machine learning, a real time waste classification algorithm can be made for the waste management system.

3.3.4 Distance Sensor

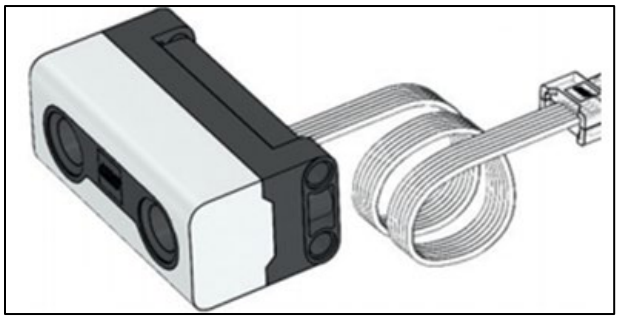


Figure 14: Spike Prime Distance Sensor [23]

Figure 14 shows the Spike Prime distance sensor. This object reads the distance of the nearest object. The distance sensor reports the nearest object in millimeters. The distance sensor will be used to avoid obstacles.

3.3.6 DC Motor

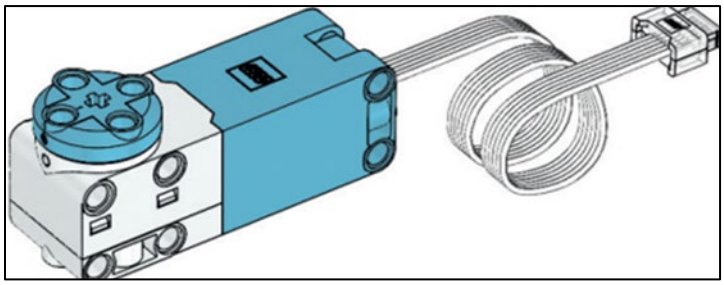


Figure 15: Spike Prime DC motor [23]

The DC motor converts electrical energy into mechanical motion. The DC motor will be used for the actuation of the waste management system. It will be used for wheels, arms, and gripper movement.

3.4 System Architecture

Figure 16 below shows a block diagram which includes all the components in the waste management system. The block diagram provides an overview of how each component communicates to the hub. The components send information to the hub, which it utilizes to act. Additionally, the hub powers the entire system.

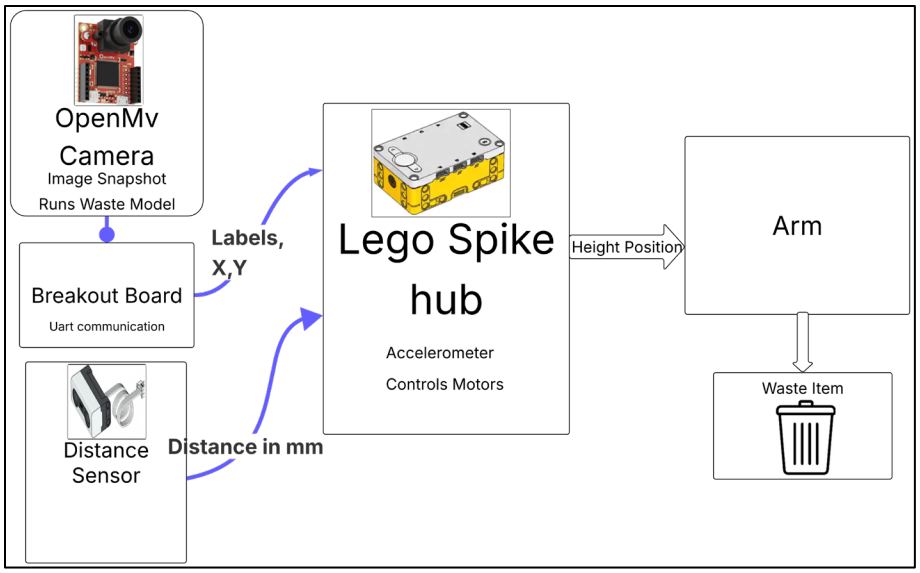


Figure 16: Block Diagram Waste Management System

As the driving base is in motion, the distance sensor will send an integer value of the nearest object or obstacle in millimeters to the Lego hub. Simultaneously, the OpenMV camera will scan its surroundings for waste. In this roaming mode, sensors send information to the hub. Then the hub decides to change course, if the distance sensor picks up an object that the camera does not deem waste.

Once waste is detected, the camera will send the classification and coordinates of the waste to the hub. The driving base then approaches the waste coordinates. The distance sensor records the distance, which the uses to determine grabbing distance Once the waste is grasped by grippers, the force sensor will confirm it with an integer value greater

than zero. The camera then scans for the corresponding bins and sends coordinates of said bin to the hub, which then transports the waste.

Overall. The Spike prime hub powers the entire system. The distance, vision, and force sensors send information to the hub. The hub will use said information for precision performance.

3.5 Robot Control Design

The flow chart in Figure 17 below details the controls design for the waste management system. The system contains five modes, which include idle, roam, target, transport, and drop off mode. The flow chart's goal is to detail how to meet criteria of all functional requirements of the system.

1. **Idle Mode** – System remains idle until activated
2. **Roaming Mode** – System keeps moving whilst dodging obstacle until waste is detected
3. **Target Mode**- System approaches, classifies, and grabs waste
4. **Transport mode** - System scans for and transports nearby bins
5. **Dropoff Mode –** System identifies corresponding bin and releases said waste into bin

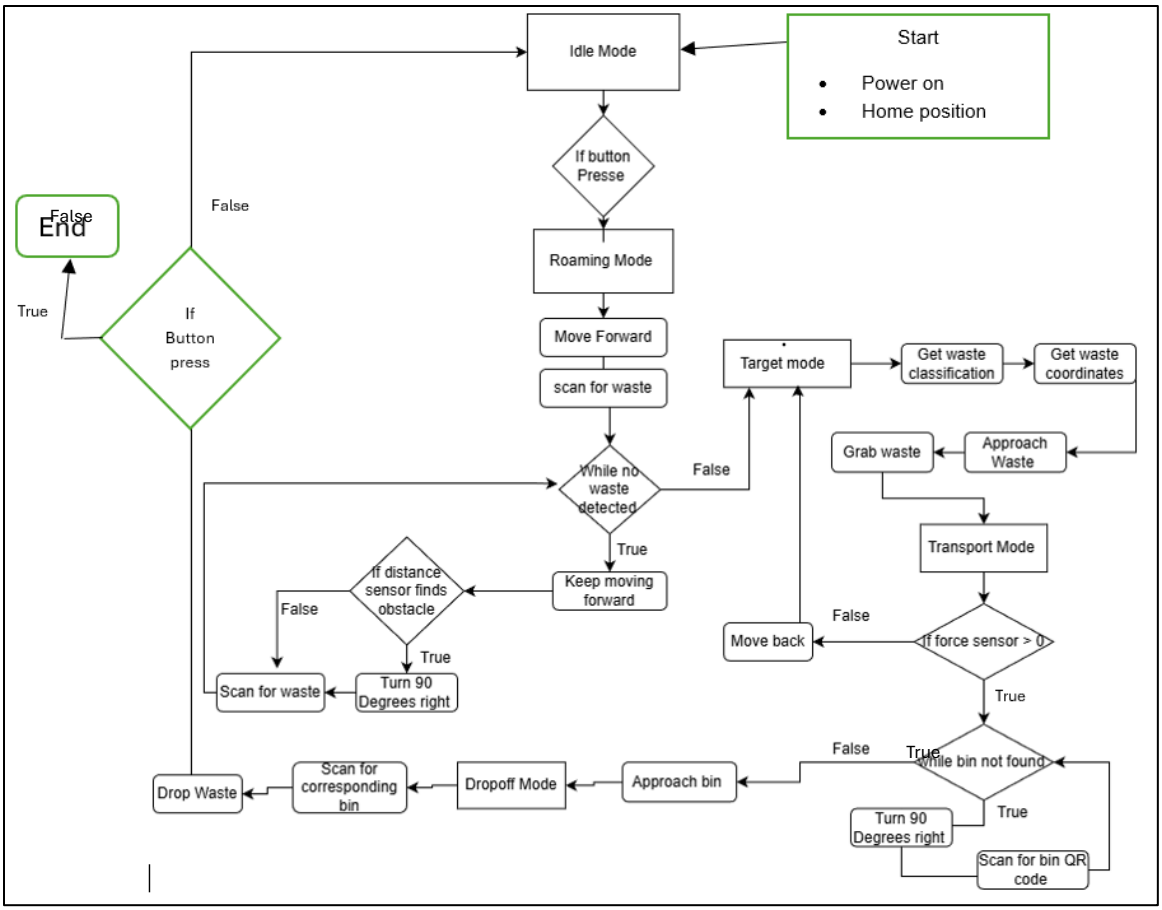


Figure 17: Flowchart for Waste Management System

3.6 Assembly Methodology

The robot is assembled using Lego spike beams, connectors, and plates for the frame and arm. Mobility is provided by two motors attached to wheels. Additional motors are dedicated to the arm's movement, at least one for basic articulation and another for gripper operation. Power is conveyed using gears and axles where necessary. A Lego Powered SPIKE controller provides centralized control, powered by a battery pack or rechargeable power source. The arm is mounted on the driving base, while also avoiding the obstruction of the sensor’s view. Due to this, sensors are mounted strategically: for example, the force sensors near the gripper, and the vision sensor at an elevated position.

Careful cable management prevents tangling or interference with moving parts. Sensor cables are routed to the controller's hub.

3.7 Testing Methodology

The system shall undergo a series of tests based on stated design requirements. First, the camera will have multiple trials on its waste classification. Noting accurate and inaccurate classifications provides insight into improving accuracy. Additionally, tests will need to be taken on how often the system can successfully transport waste to corresponding bins. Objects of different shapes and weight will also be tested on the arm. It is also important to test the robot in different conditions regarding lighting, complexity, surface, and many more. Finally, the percentage of successes in each criterion will be gathered to gauge overall performance.

Another approach to testing the systems includes using a confusion matrix: a tool for evaluating the performance of a machine learning algorithm. The structure of a confusion matrix is shown below [25].

- **True Positive (TP)** - Model predicted the positive class.

  (Predicted paper → Actual paper)

- **True Negative (TN)** - Model correctly predicted the negative class.

  (Predicted not metal → not metal)

- **False Positive (FP)** - Model predicted the positive class.

  (Predicted paper → Actual metal)

- **False Negative (FN)** - Model incorrectly predicted the negative class.

  (Predicted not metal → Actual metal**)**

Table 3: Testing Methodology Definitions

Accuracy	Measures success rate	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	A measure corrects positive of prediction	$\frac{TP}{TP + FP}$
Specificity	Ability to detect negatives	$\frac{TN}{TN + FP}$
F1 Score	Measures how well model balances to classes	$2 \times \frac{Precision \times Recall}{Precision + Recall}$
Sensitivity	Ability to detect negatives	$\frac{TP}{TP + FN}$

The data gathered from these tests is then used to formulate results on accuracy, precision, sensitivity, recall, F1 Score, and specificity. With the sensitivity and specificity gathered, a Receiver Operating Characteristic (ROC) curve can be made. The ROC curve

is a graphical tool developed to visualize and analyze the tradeoff between sensitivity and specificity [26]. The X-axis plots the False Positive Rate meaning all false positive and the Y-axis plots sensitivity. An ROC curve closer to the top left indicates higher accuracy for the waste management system.

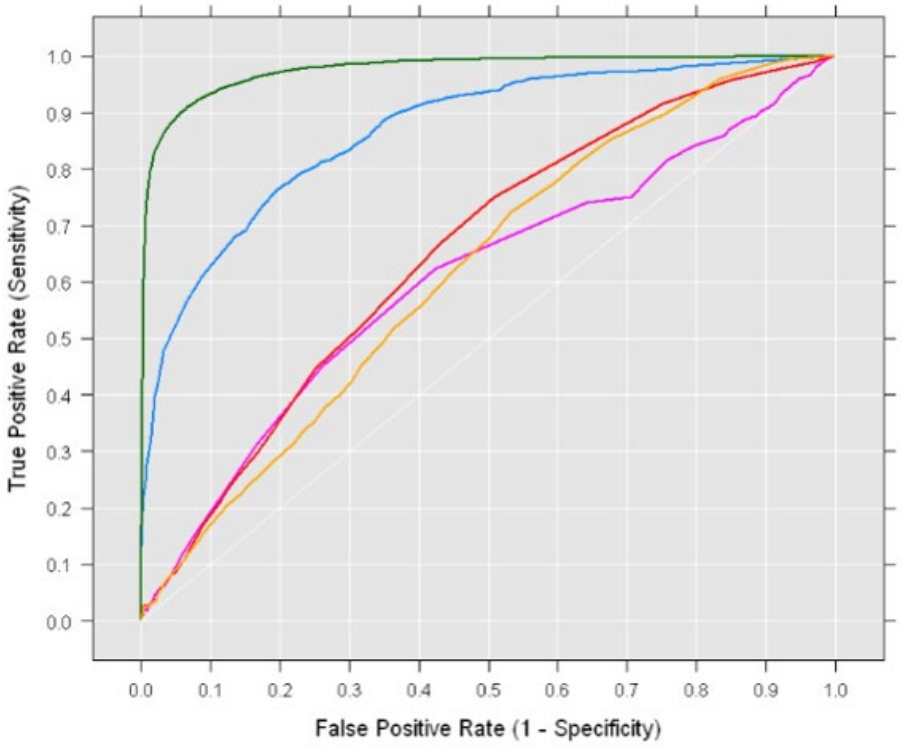


Figure 18: ROC Curve [26]

With the use of a confusion matrix the performance of the waste classification algorithm can be measured, analyzed, and visualized.

3.8 Results

It is important to analyze and present the test data of the waste management system. This includes communicating whether design requirements for the system are met. This includes discussing system successes and failures. For example, an analysis of the percentage of

successful classification of different materials provides insight into necessary steps for improving results. Results from tests in unintended conditions can also fuel future projects and enhancements to the system. Overall, communicating the results of the project reveals the project's goal and if they are satisfied.

CHAPTER FOUR: FINAL DESIGN AND ANALYSIS

4.1 Design Objective

To develop a vision-guided waste detection system, A machine learning model is trained and deployed to the OpenMV camera. This allows a LEGO-based mobile robot to identify waste objects in its environment and react accordingly in real time to transport said waste. The criteria for success in this robotics system are defined through its ability to identify, classify, and transport different waste types to their designated bins in a controlled environment.

4.2 Hardware Design

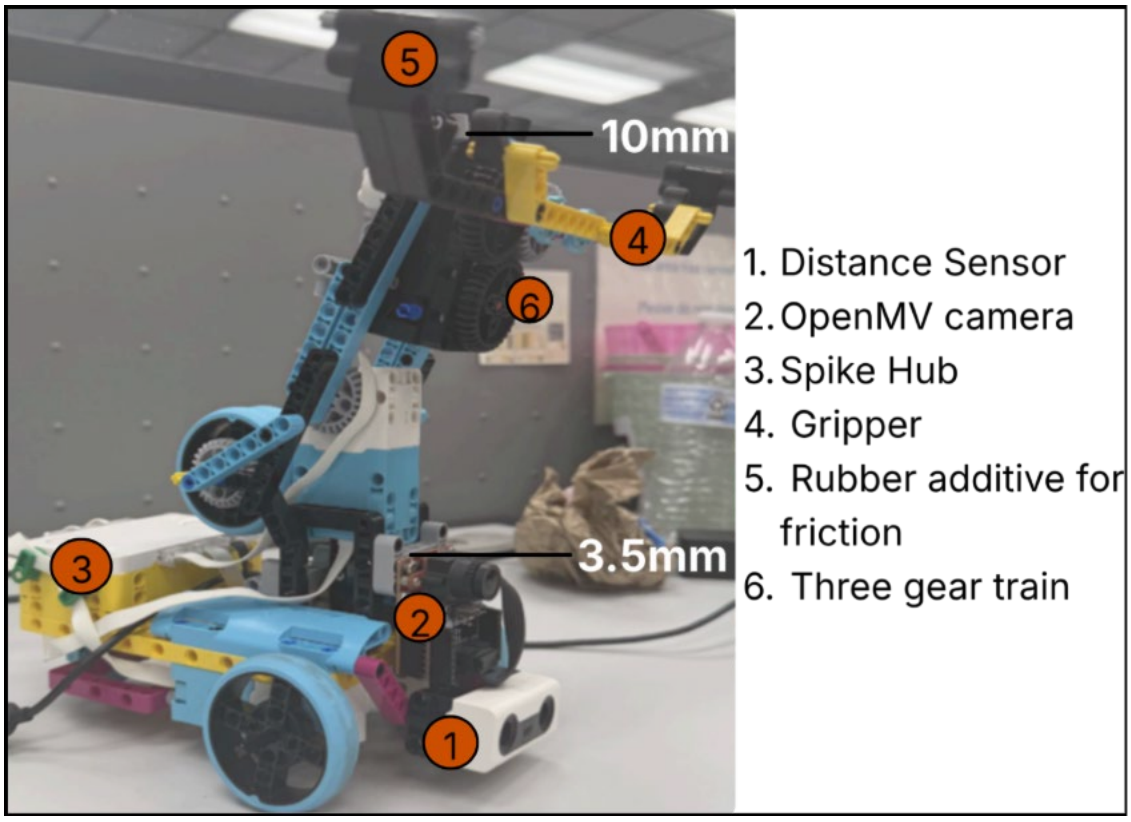


Figure 19: Final Design

The final design of the robotics system is shown in Figure 19. Key changes from the initial conceptual design include placement of the arm, camera, and distance sensor. The arm is placed in the center of the system to account for weight distribution. Located 3.5 mm from the ground at the very front, the camera is in the perfect position for detecting and targeting waste items. The same is true for the distance sensor, as its placement is very low to the ground to ensure the system never misses obstacles and objects in its path.

The gripper uses a three-gear train driven by a single motor. The first gear is attached to the motor and drives the middle gear. The middle gear then drives the third gear on the opposite side of the gripper. The middle gear acts as an idler gear, transferring motion between the two sides of the gripper. This allows both sides of the gripper to move together when the motor rotates, so the gripper can open and close using only one motor.

4.2.2 Arm Design

To ensure the structural integrity of the arm during transportation, the forces acting on it are calculated when holding a water bottle. Through calculating the forces acting along the 7 mm link of the arm, the center of mass is derived.

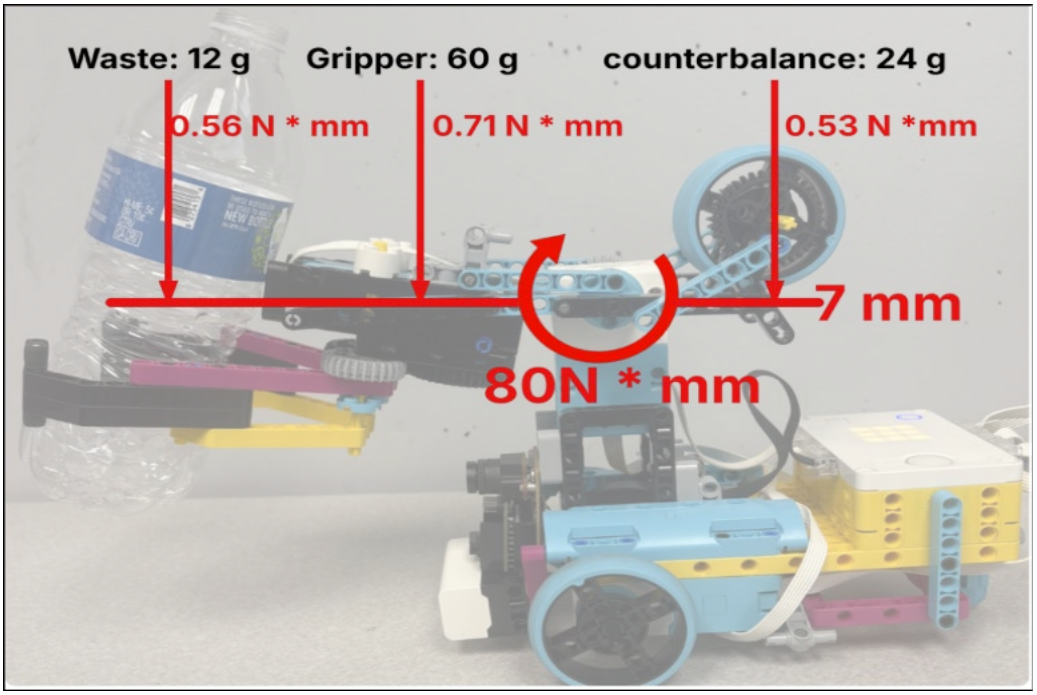


Figure 20: Robot arm structure

$$\text{Center of Mass} = \frac{(12g \cdot 0)+(60g \cdot 3.55mm)+(24 \cdot 7mm)}{12g+60g+24g} = 3.94mm$$

With the center of mass defined, the total downward torque acting on the arm link is calculated as, $\tau = F \cdot d$, where d is the distance from the center of mass.

$$\tau_{\text{Gripper}} = 0.59N \bullet 1.2\ mm = 0.71\ N \bullet mm$$

$$\tau_{\text{Bottle}} = 0.12N \bullet 4.7\ mm = 0.56\ N \bullet mm$$

$$\tau_{\text{Counterweight}} = 0.59N \bullet 1.59mm = 0.53\ N \bullet mm$$

$$\text{Total torque at rest} = (0.71 + 0.56) - 0.53 = 0.74\ N \bullet mm$$

Whilst holding a water bottle, the total downward torque acting on the 7 mm arm link when the motor is free spinning is 0.74 Newton millimeters. However, the Spike prime motor boasts a normal operating torque of 80 Newton millimeters. These calculations reassure

the fact that the Waste bot can transport waste items within and above the weight of a water bottle.

4.3 Software and Control Logic

4.3.1 Model Training

One of the Waste bot's main functions is to detect waste types in real time, which requires a machine learning model. To develop a waste detection model, the platform used was Edge impulse. It allows for a simple training process that involves uploading pictures, adding label bounding boxes, and setting training parameters. Edge impulse then provides a folder containing example code and a TensorFlow lite file for deploying to the Openmv. In total, over six thousand pictures of waste objects were gathered through pictures and online datasets [27] [28] [29]. Figure 21 below provides a glimpse of the training process and a visualization of the different waste classes clustered by similarity. The paper and plastic clusters are distinguishable, but the same cannot be said for metal.

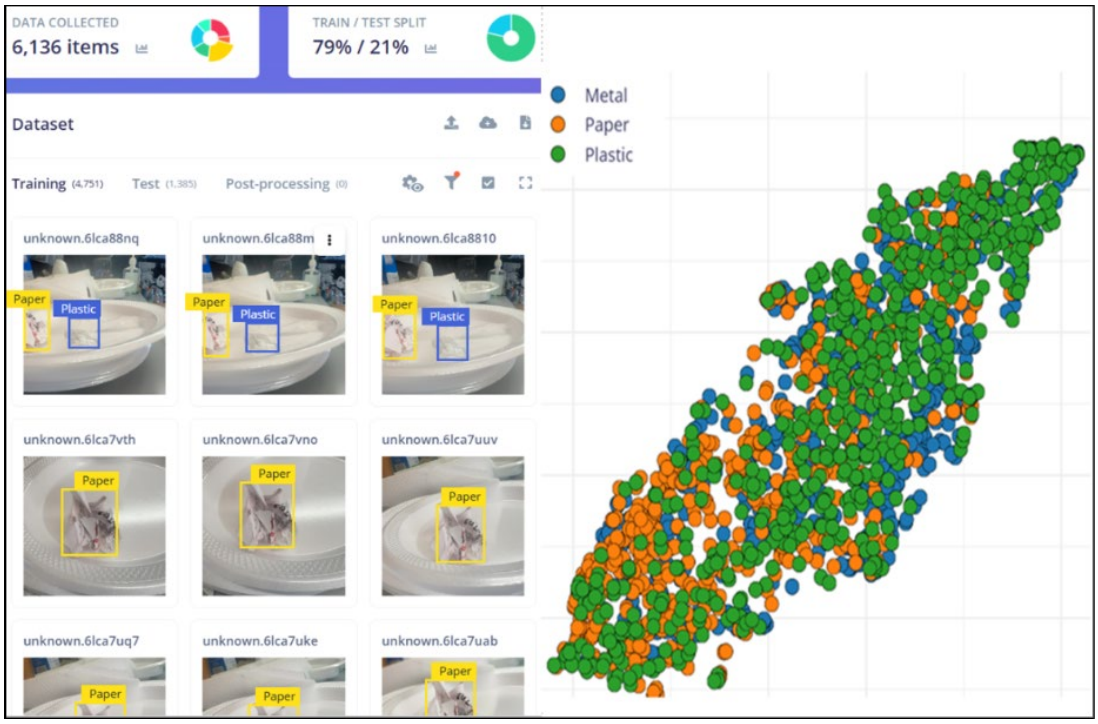


Figure 21: Model training process

4.4 Testing Methodology

Testing for the Waste bot is narrowed down to its ability to classify waste, as it has already proven successful in roaming, targeting, and transporting waste. First, a test set of 20 waste trials will be prepared for each class. The cameras' prediction of each waste item in the plastic, metal, and paper test set is recorded. Then, from the gathered data, a confusion matrix is built and key performance metrics such as accuracy, precision, f1 score, sensitivity, and specificity will be calculated for analysis.

4.5 Results and Analysis

Table 4 below is a confusion matrix which compares the Waste bot's predictions on each test set to the actual correct label. The rows represent the Waste bot predicted class, while the columns represent the actual class. Diagonal cells of the table are the correct

predictions. Since the test for a waste class was 20 items, each column has a sum of 20 items.

Table 4: Confusion Matrix of results

		Predicted		
		Paper	Metal	Plastic
Actual	Paper	**18**	0	2
	Metal	0	**12**	8
	Plastic	0	1	**19**

Table 4 reveals that paper and plastic detections for the Waste bot are high in accuracy as they've only missed at most 2 detections. However, the same cannot be said for metal detection which only got about half right. Almost half of metal objects were detected as plastic. This may be due to lighting conditions which give metal objects a glossy appearance resembling plastic. This is further supported by the edge impulse feature explorer shown in Figure 21, in which there is barely a visible metal cluster as it lays under the plastic cluster.

The performance of the classification model was evaluated using accuracy, precision, recall, and F1 score. The overall accuracy of the model was 81.7%, indicating that most predictions were correct. The model performed very well on the paper class, achieving a precision of 1.00 and a recall of 0.90, meaning it correctly identified most paper items without making false positive errors. For plastic, the recall was high at 0.95, showing that most plastic items were detected, but the precision was lower at 0.66, indicating that some non-plastic items were incorrectly classified as plastic. The metal class showed a lower recall of 0.60, meaning that a large number of metal items were missed, although its precision was high at 0.92. This suggests that while the model is accurate when predicting metal, it does not detect all metal instances effectively. The F1 scores reflect this balance, with paper achieving the highest score (0.95), followed by plastic (0.78) and metal (0.73). Overall, the model performs well but could be improved by increasing recall for the metal class and precision for the plastic class.

Table 5: Performance Metrics

Metric	Description	Formula	Paper	Metal	Plastic
Precision	A measure corrects positive of prediction	$\frac{TP}{TP+FP}$	1.00	0.92	0.65
Specificity	A measure of negatives found	$\frac{TN}{TN+FP}$	1.00	0.97	0.75
Recall	A measure of actual positives found	$\frac{TN}{TN+FN}$	0.90	0.60	0.95
F1 Score	Measures how well model balances each class	$2 * \frac{Precision \cdot Recall}{Precision + Recall}$	0.95	0.73	0.78
Accuracy	Measures success rate	$\frac{TP+TN}{TP+TN+FP+FN}$	$\frac{18+19+12}{60} = 0.81$		

CHAPTER 5: CONCLUSIONS AND FUTURE IMPROVEMENTS

5.1 Conclusion

This project has proven successful in designing, building, and programing a mobile pick and place robot capable of detecting, classifying, and transporting waste. Within a controlled environment, the robot was able to navigate, identify and sort different waste materials, using the OpenMV camera for classification and the LEGO-based gripper mechanism for physical interaction.

The robot gripper was successfully designed to pick up and hold lightweight objects using a single motor and a gear-driven mechanism. The use of a three-gear system allowed both sides of the gripper to move together, enabling efficient opening and closing while keeping the design simple. The inclusion of a counterbalance helped reduce the torque required from the motor, improving stability, and performance.

The classification model used to identify materials achieved an overall accuracy of 81.7%. The model performed best on the paper class, with high precision and recall, while performance on the metal class was limited by lower recall, indicating missed detections. The plastic class showed high recall but lower precision and specificity, suggesting that the model tended to over-predict plastic. These results highlight areas for improvement, particularly in better distinguishing between metal and plastic.

Overall, the system met its design objectives, combining effective mechanical design with a reasonably accurate classification model.

5.2 Future Improvements

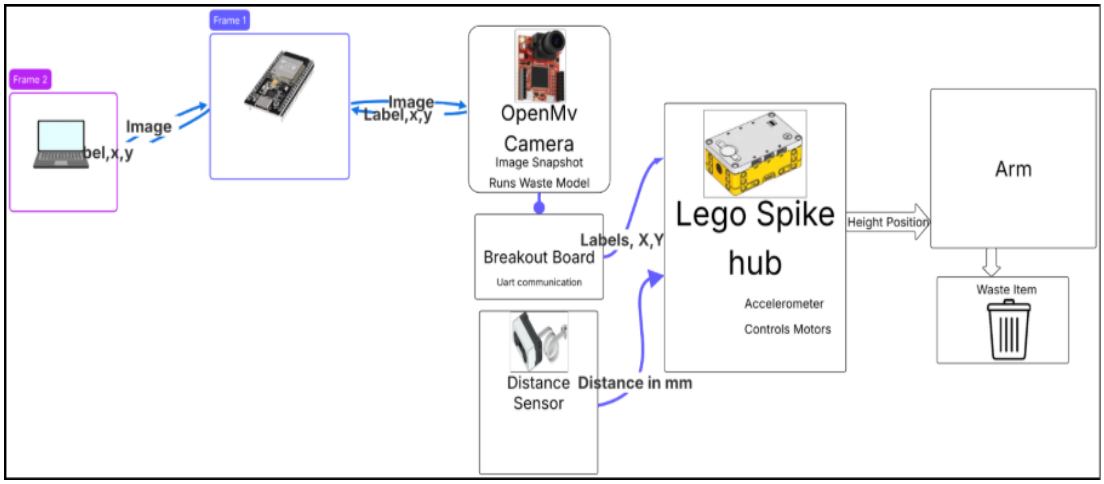


Figure 22: Improved Architecture

Future improvements for this prototype would involve departing from the Lego Spike Prime kit. This would involve using sturdier building material, and stronger motors. The use of a microcontroller instead allows for more sensor inclusion and compute power. This would result in a more sophisticated navigation method. Additionally, the waste detection and classification accuracy could be improved if a stronger machine learning model were running on a separate computer. As shown in Figure 22, images would be sent to a PC server, which would perform detections and send results back to the system. could include optimizing the balance of the arm, refining the gear ratios for smoother control, and improving the classification model to increase recall for metal and precision for plastic.

APPENDIXES

Appendix A: Issue Report 1

Project Issue 1:

Detailed discussion of each realistic constraint, which follows, as they are applied to the design/technical project: safety; security; reliability; aesthetics; ethics; power requirements; size and weight

Answering Project Issue:

In our Capstone project, the goal is to build a prototype pick and place robot with machine vision to handle waste management. Multiple constraints presented themselves during development of the system. For example, constraints relating to budget, weight, power, skill, reliability, and sanitary issues were discovered.

First, financial constraints limit development to the available resources. Therefore, the waste management robot must compromise between budget in terms of its capabilities. For assembling the bulk of the robot, the Lego Spike Prime Kit is utilized. The Lego Spike Prime kit provides beams, gears, platforms, connectors, and wheels in Lego plastic, which allows for a robust, stable frame. In addition, the kit provides sensors, motors, and a hub for simple connection and manipulation. Although Lego provides a cheaper alternative for building a robot, a fully automated system cannot be achieved without utilizing third-party components such as the OpenMV camera. However, to best utilize this vision sensor, a robust, accurate machine learning algorithm must be built for waste detection and classification. Unfortunately, financial constraints are limiting the systems' waste

recognition accuracy. This is due to hefty subscription requirements from online platforms to utilize the best tool for creating an object detection algorithm. In our case, the labeling of large datasets of waste is locked behind a Google Gemini subscription. This roadblock resulted in a long, inaccurate labeling of waste datasets, therefore impacting the accuracy of the waste classification algorithm. Overall, financial constraints have limited the resources used to build and accurately control the robotic system.

There are many constraints that restrict the Lego robotic system from being applicable to certain real-world environments. Considering current equipment, outdoor environments may prove difficult to navigate as there are many unpredictable outcomes such as terrain and lighting. Additionally, the current system does not have the capability to detect and transport waste objects located on an elevated platform.

Although the robotic system is dealing with waste, the contaminating aspect of waste has not been considered. Unfortunately, there are no functions for dealing with contamination of the system whilst transporting waste. Therefore, the robotic system could prove to be unsanitary when dealing with unhygienic materials.

Legos material is not intended for carrying much mass; therefore, moving objects heavier than the system could pose a potential threat. As a result, adding enough support to not only the base but also the arm itself, could negate issues of tackling heavier objects. To further account for balance, the system is restricted to transporting items of 100 grams or less.

Additionally, the gripper for the system is assembled with Lego parts. When lifting certain objects, they tend to slip as the materials and parts provided do not provide a strong grip. Therefore, the size of objects transported must have a width of 6 to 14 centimeters. The height of objects must also be above 4 centimeters.

In terms of power, the Lego spike hub contains a battery of 2100mah. However, accounting for the powering of the actuator, sensor, and third-party systems, the waste management system may require frequent charging and a short battery life.

The computer science skills required for localization and advanced machine learning are lacking. This makes it difficult for the robot to map its environment and memorize bin locations for waste transport. Alternatively, fiducial tags are used to represent bin locations as they produce signal instructions the system can respond to. With the current skills, a fully autonomous system may be impossible, but there are shortcuts to providing results.

Appendix B: Issue Report 2

Project Issue 2:

Discussion of the impact of your project considering contemporary and social issues

Answering the project issue:

Waste Pollution is a global crisis created through the mismanagement of waste. With an ever-growing human population, there is a rapid increase in the generation of garbage, as shown through the two billion tons of waste generated annually. A major solution to reducing waste pollution requires the reuse of materials rather than dumping them at landfills. Unfortunately, most garbage disposed of by landfills or burning contributes to climate change, poor respiratory health, and human and wildlife endangerment. The uncontrollable waste problem creates many pollutants, which not only affects the environment, but also life on earth.

Another factor to consider is how poor waste management affects biodiversity. Uncontrollable waste has led to the displacement of wildlife. In addition, chemical plastic pollutants seep into environments, which harm wildlife as they are ingested. Later, these same plastics are transported to and found in human organs.

Individuals are burdened with the responsibility of reducing their waste production while also correctly recycling their waste. However, tackling plastic pollution on the individual level has not proved effective as there is little concern and any efforts made are inconvenient. Additionally, efforts may not produce results as 25% of waste in recycling bins ends up thrown away due to human error [1]. This is due to improper disposal or

classification of waste which leads to contamination of recyclable material. As things are now, human error will remain a contributing factor to the two-thirds of waste that goes unrecycled in the United States [1].

It is for these reasons that robotics should be utilized to assist in waste management. With robotics, accurate classification and collection of recyclable materials is automated, therefore reducing contamination of recyclables. The mission of this project is to design a robotic system that actively roams and collects recyclable waste in public and living spaces. The intention of designing a waste management robot is to attempt to demonstrate the use of a machine in reducing the buildup of global waste. Through this demonstration, a large-scale implementation of waste collecting robots deployed in public spaces could prove successful and reduce waste pollution. Overall, the project aligns with the rising initiatives of circular economies, sustainable cities, and climate action.

Utilizing robotics for waste management aligns with the global trend of smart cities. Smart cities are urban areas that use technology, data, and governance to enhance sustainability, efficiency, and quality of life for its residents. Similarly to the goal of the waste management robot, smart cities aim to reduce resource consumption. The utilization of automated waste management on the scale of a city could prove useful in sustainability.

Additionally, the design of a waste management robot could act as a solution to health and safety, especially in traditional waste handling workspaces. Employees of such

workspaces are usually exposed to hazardous materials whilst handling physical demanding tasks. As an alternative to manual waste work, a waste management robot protects from any potential employee shortages while also reducing human errors and potential injuries.

In conclusion, waste pollution is a serious and escalating global issue driven by mismanagement and human error. Traditional recycling efforts alone have proven insufficient to address the scale of the problem. Implementing robotic waste management systems offers a promising solution by improving recycling accuracy, reducing environmental harm, supporting smart city development, and enhancing worker safety.

Appendix C: Project Issue Report 3

Discussion of the feasibility of implementing your project from a business and economic perspective.

Answering Project Issue:

Proper waste management is a growing market due to rising environmental concern and population growth, especially in urban environments. With labor costs and worker shortages to consider, automation becomes more effective than manual labor in waste management. Existing commercial products such as AI-powered sorting systems, smart bins, and industrial sorters confirm the technical and commercial feasibility of automating waste management at various scales. These factors illustrate the business viability for a waste management robot that detects, classifies, and transport waste in its environment.

Our current project, Waste bot is assembled using LEGO Spike Prime hardware and an OpenMV camera for computer vision, which runs a FOMO object detection model trained through Edge Impulse. Waste bot navigates its environment to detect and transport waste to designated points, which are identified using AprilTag QR codes. It is a prototype with battery life constraints, limited payload size and weight, and varied accuracy dependent on lighting. The system's navigation methods rely on the ultrasonic distance sensor, AprilTags, and a record of movements mapped to an 2D chart. These restrictions hinder the Waste bot from succeeding in complex, real-world environments. Unfortunately, the Waste bot in its current form cannot be implemented into real world environments and is therefore not commercially viable.

Implementing a usable product requires replacing Lego materials and hardware with custom material that is durable, which also provides an opportunity to increase battery life.

Instead of using AprilTags, a truly viable product would likely need SLAM (Simultaneous Localization and Mapping) for navigation. SLAM utilizes sensor data such as LIDAR to map objects in an environment relative to the robot's body. In addition to this, upgrading to a camera with better processing power allows for a more complex and therefore accurate machine learning model to detect waste.

The Wastebot prototype is not equipped for real world implementation, but it can be utilized as an educational tool. If utilized as a kit, the Wastebot could provide an introduction into robotics, machine learning, and computer vision. As an educational kit, the Wastebot would include Lego Spike Prime hardware, the OpenMV camera, and a pretrained waste detection model. With a guide provided to building a waste management system, students and hobbyists can gain experience in covering Python programming, and machine learning concepts.

While the Wastebot prototype can serve as an educational guide, its transition to a commercially viable business depends on upgrading current hardware to industrial-grade hardware.

Appendix D: Project Issue Report 4

Discussion of how ASME, OSHA and other codes and standards impact the project.

Answering Project Issue:

The design and implementation of a management robot is influenced by several engineering codes and safety standards, particularly those from Occupational Safety and Health Administration (OSHA), and other relevant organizations. These standards help ensure that the robot is safe, reliable, and suitable for real-world deployment.

ASME Standards

Although the Waste bot is assembled with LEGO SPIKE Prime components, ASME principles still provide a guideline for achieving an efficient system. This resulted in considerations such as mechanical integrity and design safety during the development process. This is especially true regarding the Waste bot's arm. The design of each arm link required careful consideration of the forces acting on the motors connecting them. This is ensuring that the Lego motors could support the weight of the link and of the waste being transported. With the addition of a counterbalance at the arm's joint. The system would prove structural integrity after multiple test trials.

OSHA Regulations

With safety as a priority, OSHA standards are critical if the Wastebot is deployed in human environments. For the Wastebot, this means limiting the movement speed

to 130 mm/s. Another safety requirement is the addition of emergency stops. Henceforth, distance, motor, and gyro sensors are utilized to measure and react to danger conditions. These emergency conditions include if something is detected less than 80 millimeters away, if the system motors are stalled, or if the systems roll and pitch becomes higher than 15 degrees. Should any of these conditions come true. The WasteBot halts every movement and waits for a designated time before retreating from the premises. Additionally, hazardous waste types such as glass are left out of the classification due to this Waste Bot being only a prototype.

Environmental Regulations

The Environmental Protection Agency (EPA) sets standards that relate directly to proper waste sorting categories, including metal, plastic, and paper. These standards help to determine which waste is considered non-hazardous and recyclable. Additionally, these standards help group waste types, set up testing criteria, and avoid contamination.

ASME, OSHA, and EPA standards play a crucial role in shaping the waste management robot by ensuring it is structurally intact, safe for human interaction, and environmentally responsible. As a result, the final design of the Waste bot contains controlled waste interaction, multiple safety precautions, speed limitations, and even weight distribution. Even though the project is educational and built with LEGO SPIKE Prime and OpenMV, applying these

professional standards prepares the system for potential real-world applications and improves its overall design quality.

Appendix E: Project Issue Report 5

Explain, in detail, how you plan to remain technologically current in your field over the life of your employment

Answering Project Issue:

The creation of automated systems is an exciting field which is constantly evolving. The field changes quickly, especially in areas such as robotics, artificial intelligence, computer vision, and automation. Which is why staying current helps to remain effective, safe, employable, and solve problems. Getting into the field of embedded systems and robotics requires broad skills that involve programming, electronics, mechanics, and an understanding of artificial intelligence. My goal is to become a viable employee in this field, through obtaining industry certifications

Professional Development

One key step to becoming effective for employers is to obtain certifications relevant to industry standards. For this reason, I will be pursuing Siemens or Allen-Bradley PLC programming certificates.

I intend to obtain certifications for python and C++. These programming languages are common for embedded systems. The main certifications to attempt will be PCEP and C++ institute certification exams. Another certification I intend to obtain is a machine learning certificate through a course in Coursera. These certifications are important because mechatronics heavily involve coding and AI.

One of the fundamental skills required for robotics design is Computer aided design or CAD. Which is unfortunately one of my weaknesses as I've had little to no experience with it. I hope to also add this to my list of skills eventually.

In addition to certifications, gaining hands-on experience is essential for developing practical skills in robotics and embedded systems. I plan to continue working on personal and academic projects, such as building robotic systems that integrate sensors, actuators, and computer vision. Projects like my LEGO SPIKE Prime and OpenMV waste management robot demonstrate my ability to apply programming, electronics, and mechanical design in a real-world context.

Through these projects, I will continue improving my ability to debug systems, optimize performance, and design reliable solutions. I also aim to expand my experience by learning new platforms such as Arduino and Raspberry Pi, which are widely used in embedded systems.

Continuous Learning

To remain technologically current throughout my employment, I plan to make continuous learning a habit in my career. This will be done through attending conferences, reading documentation, journals, case studies, and pursuing personal projects. For example, a developing technique for robotics control is the use of a vision language action model (VLA). In short, these models combine vision with a prompt, which the robot can interpret and react accordingly to its environment. This technique could bridge the gap of possibility

for robots, as they could operate in any environment. It is papers like these that I intend to consume to keep an up-to-date understanding of the field.

In conclusion, remaining technologically current in the field of robotics and embedded systems requires a combination of continuous learning, practical experience, and professional development. I will continue developing my skills through hands-on projects, collaboration, and exposure to real-world applications.

Appendix F: Spike Prime Hub Code

```
from pybricks.hubs import PrimeHub
from pybricks.pupdevices import Motor, UltrasonicSensor
from pybricks.parameters import Button, Direction, Port, Side, Stop
from pybricks.robotics import DriveBase
from pybricks.tools import wait, StopWatch
from pupremote_hub import PUPRemoteHub
import umath, urandom

#Defining Variables
rand = 1 if urandom.randint(0, 1) else -1
material = 0
#ports
sw= StopWatch()
p = PUPRemoteHub(Port.E)
p.add_command('msg',"repr","repr")
p.add_channel('one_w',to_hub_fmt='bbbb')
p.add_command('num', from_hub_fmt="b", to_hub_fmt="b")
gripper = Motor(Port.A, gears =[36,28])
LeverArm  = Motor(Port.D)
LDB = Motor(Port.F, Direction.COUNTERCLOCKWISE)
RDB = Motor(Port.B)
LDB.control.stall_tolerances(speed=130, time=30)
RDB.control.stall_tolerances(speed=130, time=30)
DS = UltrasonicSensor(Port.C)
hub = PrimeHub()
lastxpos= False
drive = DriveBase(LDB, RDB, wheel_diameter=56, axle_track=130)
hub.display.orientation(Side.LEFT)

print(hub.imu.heading(), "degrees")

label = [ 'metal','aper', 'plastic']
Start = False
mode = 0
runs = 0
detected = 0
dlabel = 0
xpos = 0
hpos = 0
dOBJ = 2000
xychart = [0, 0]
centered = False
GrabbingDist = False
category = [0, 0, 0, 1,2,2]
lastexpos = False
```

Appendix G: OpenMV Code

```
import sensor, image, time, ml, math, tf
from pupremote import PUPRemoteSensor
from pyb import LED
from ml.utils import NMS
msg = ''
dlabel = 0
xpos = 0
hpos = 0
mode = 0
num = 0
cx = 0
cy = 0
Lastxpos = False
istag = False
# ── Per-class confidence thresholds ──────────────────────────
CLASS_CONFIDENCE = {
'background'        :0.9,
 'Metal can'        :0.79,
 'Metal foil'       :0.68,
 'Paper'            :0.60,
 'Plastic'          :0.73,
 'Plastic bottle'   :0.73
}

# ── Detection tuning ─────────────────────────────────────────
MIN_CONFIDENCE = min(CLASS_CONFIDENCE.values())
def num(n):
    global mode
    mode = n
    return n
# ── PUPRemote Setup (LEGO Spike communication) ───────────────
p = PUPRemoteSensor(power=False)
p.add_command('msg', 'repr', 'repr')
p.add_channel('one_w', to_hub_fmt='bbbb')   # sends int to hub (1=detected, 0=not)
p.add_command('num',from_hub_fmt='b', to_hub_fmt='b')

# ── LED indicator ────────────────────────────────────────────
led = LED(2)
rled = LED(1)
led.on()


# raise to cut false positives (0.0 - 1.0)
MIN_SIZE         = 30   # minimum blob width/height in pixels
MAX_SIZE         = 320  # maximum blob width/height in pixels
MIN_ASPECT       = 0.2 # minimum width/height ratio (filters flat blobs)
MAX_ASPECT       = 30  # maximum width/height ratio
STREAK_NEEDED    = 2 # frames object must appear before confirming detection
ignoreColor = [(68,86,-31,-10,2,22),(47,75,27,56,-21,5)]

# ── Camera Setup ─────────────────────────────────────────────
sensor.reset()
sensor.set_pixformat(sensor.RGB565)
sensor.set_framesize(sensor.QQVGA)
sensor.set_windowing((160,160))
sensor.set_auto_whitebal(False)
sensor.set_auto_gain(False)
sensor.set_auto_exposure(False, exposure_us=10000)
sensor.skip_frames(time=2000)
#sensor.set_hmirror(True)    # ← flip horizontal
#sensor.set_vflip(True)


# ── Load Model ───────────────────────────────────────────────
net    = None
labels = None

try:
    net = ml.Model("trained.tflite", load_to_fb= False)

except Exception as e:
    raise Exception(
        'Failed to load "trained.tflite" — '
        'did you copy it to the OpenMV drive? (' + str(e) + ')'
    )

try:
    labels = [line.rstrip('\n') for line in open("labels.txt")]
except Exception as e:
    raise Exception(
        'Failed to load "labels.txt" — '
        'did you copy it to the OpenMV drive? (' + str(e) + ')'
    )

print("Model loaded. Classes:", labels)
```

Resume

Ndiouga Niang

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EDUCATION

Tennessee State University, Nashville **2026**

Bachelor of Science in Mechatronics Technology
Minor: Computer Science
Expected Graduation: May 2026
GPA: 3.39 / 4.0
Relevant course work: Basic Electronics, Circuits and Devices, Intro to Robotics, Calculus I, Python Intro to microprocessors, Data comm and Computer Networks

TECHNICAL PROJECTS

Vision enabled Waste Classification Robot Using Openmv and Lego Hardware

- Designed and developed an autonomous waste-sorting robot using LEGO Spike Prime, OpenMV camera, and VL53L0X time-of-flight sensor for navigation and waste detection, classification, and transportation

-Used Edge Impulse to Train a custom FOMO MobileNetV2 object detection model on the TACO dataset (1,500+ images, 5 classes)

-Implemented computer vision algorithms in MicroPython to classify and sort waste objects in real time using edge AI on embedded hardware

Fluid Powered Vehicle Challenge

- Collaborated in a team to design and analyze a human-powered vehicle using hydraulic and pneumatic systems
- Evaluated and optimized hydraulic circuit performance for competition conditions
- Diagnosed and resolved mechanical and system-level issues to improve reliability and performance

WORK EXPERIENCE

Walmart, Memphis Tn **2022- Present**

Meat and Produce Associate

- Package and label meat products
- Stock backrooms with meat products
- Assist customers

McDonalds, Olive Branch **2019- 2021**

Team member

- Maintain Hygiene and cleanliness in the store
- Prepare customers' orders

REFERENCES

1. Moraes, Cassiano Ferro. “Pick and Place Robots: An In-Depth Guide to Their Functionality and Applications.” *Www.wevolver.com*, 25 Mar. 2024, www.wevolver.com/article/pick-and-place-robots-an-in-depth-guide-to-their-functionality-and-applications. Accessed 26 Sept. 2025.
2. Forlizzi, J., & DiSalvo, C. (2006, March). Service robots in the domestic environment: a study of the roomba vacuum in the home. In *Proceedings of the 1st ACM SIGCHI/SIGART conference on Human-robot interaction* (pp. 258-265).
3. Earthday. “How Our Trash Impacts the Environment.” *Earth Day*, 9 July 2024, www.earthday.org/how-our-trash-impacts-the-environment/. Accessed 26 Sept. 2025.
4. UNEP. “Plastic Pollution.” *UNEP - UN Environment Programme*, United Nations Environment Programme, 2025, www.unep.org/plastic-pollution. Accessed 26 Sept. 2025.
5. https://www.facebook.com/unep. (2024). Eight ways to overcome the waste pollution crisis. UNEP. https://www.unep.org/news-and-stories/story/eight-ways-overcome-waste-pollution-crisis?
6. Komane, B., & Mathonsi, T. (n.d.). Design of a Smart Waste Management System for the City of Johannesburg. Accessed September 26, 2025. https://arxiv.org/pdf/2303.14436
7. Waste Robotics. “Beyond the Bin, Designing a Future without Waste.” *Wasterobotic.com*, Waste Robotics Inc., 14 Jan. 2025, blog.wasterobotic.com/en/blog/breaking-the-waste-cycle-embracing-a-circular-future-for-business. Accessed 26 Sept. 2025.
8. OECD. “Plastic Pollution Is Growing Relentlessly as Waste Management and Recycling Fall Short, Says OECD.” *OECD*, 2024, www.oecd.org/en/about/news/press-releases/2022/02/plastic-pollution-is-growing-relentlessly-as-waste-management-and-recycling-fall-short.html?utm_source=chatgpt.com. Accessed 26 Sept. 2025.
9. “Historical Development of Robotics.” *Discover Engineering*, 24 Feb. 2025, www.discoverengineering.org/historical-development-of-robotics/ Accessed 12 Oct. 2025.
10. White, Christopher. “A Brief History of the Robot Vacuum Cleaner - Vacuum Wars.” Vacuum Wars, 4 Feb. 2023, vacuumwars.com/history-of-the-robot-vacuum-cleaner/. Accessed 14 Oct. 2025.
11. Soori, M., Arezoo, B., & Dastres, R. (2023). Artificial Intelligence, Machine Learning and Deep Learning in Advanced Robotics, A Review. *Cognitive Robotics*, *3*(1), 54–70. ScienceDirect. https://doi.org/10.1016/j.cogr.2023.04.001
12. Wasim, Hamza. “Robots in the Manufacturing Industry: Types and Applications.” *Www.wevolver.com*, 4 May 2023, http://www.wevolver.com/article/manufacturing-robots . Accessed 23 Oct 2025
13. Morgan, A. A., Abdi, J., Syed, M. A. Q., Kohen, G. E., Barlow, P., & Vizcaychipi, M. P. (2022). Robots in Healthcare: a Scoping Review. *Current robotics*

reports, *3*(4), 271–280. https://doi.org/10.1007/s43154-022-00095-4. Accessed 13 Oct 2025.
14. Forlizzi, J., & DiSalvo, C. (2006, March). Service robots in the domestic environment: a study of the roomba vacuum in the home. In *Proceedings of the 1st ACM SIGCHI/SIGART conference on Human-robot interaction* (pp. 258-265). Accessed 13 Oct 2025.
15. Patel, N., & Mehta, R. (2022). *Low-Cost Robotic Arm for Waste Sorting Using Arduino*. Journal of Engineering Innovation. Accessed 13 Oct 2025.
16. Ferrari, G., & Ferrari, M. (2002). *Building Robots with Lego Mindstorms* (1st Edition). Elsevier Science. https://shop.elsevier.com/books/building-robots-with-lego-mindstorms/ferrari/978-1-928994-67-1 . Accessed 13 Oct 2025.
17. Wong, K.-Y., Pu, S.-C., & Wong, C.-C. (2024). A Robotics Experimental Design Method Based on PDCA: A Case Study of Wall-Following Robots. *Sensors*, *24*(6), 1869. https://doi.org/10.3390/s24061869. Accessed 25 Oct. 2025
18. Y. Liu *et al*., "Mobile Robot Localisation and Navigation Using LEGO NXT and Ultrasonic Sensor," *2018 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Kuala Lumpur, Malaysia, 2018, pp. 1088-1093, doi: 10.1109/ROBIO.2018.8665350.
19. Azis, Fatin Amanina & Suhaimi, Hazwani & Abas, Pg Emeroylariffion. (2023). The Development of an Automated Waste Segregator. International Journal of Integrated Engineering. 15. 19-30. 10.30880/ijie.2023.15.04.002. Accessed 13 Oct 2025.
20. *daVinci Robot*. (n.d.).Www.unitymedcenter.com. https://www.unitymedcenter.com/hospital/davinci-surgical-system.html. Accessed October 25, 2025
21. [tug-robot-mercy-hospital-hallway-navigation]. (2025). https://www.mercy.net/about/mercy-technology-services/tug-robot/. Accessed October 25, 2025
22. iROBOT. (2025). Roomba Combo detects dirt. In *iROBOT.com*. https://www.irobot.com/en_US/roomba-combo-j7plus-robot-vacuum-and-mop/C755020.html. Accessed 25 Oct. 2025.
23. *LEGO Education SPIKE*. (n.d.). Spike.legoeducation.com. https://spike.legoeducation.com/prime/start/
24. *OpenMV4 Cam H7 R2*. (2025). Singtown.com. https://singtown.com/us/store/openmv4-h7-r2
25. GeeksforGeeks. (2025, May 30). *Understanding the confusion matrix in machine learning*. GeeksforGeeks. https://www.geeksforgeeks.org/machine-learning/confusion-matrix-machine-learning/
26. Chan, C. (2025, August 15). *What is a ROC Curve and How to Interpret It*. Displayr. https://www.displayr.com/what-is-a-roc-curve-how-to-interpret-it/
27. Proença, P. F., & Simões, P. (2020). TACO: Trash Annotations in Context for Litter Detection. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2003.06975

28. *YOLO Waste Detection Computer Vision Model*. (n.d.). [Dataset]. https://universe.roboflow.com/projectverba/yolo-waste-detection
29. Sashaank, S. (n.d.). *Waste Classification data* [Dataset]. https://www.kaggle.com/datasets/techsash/waste-classification-data/data